Abnormal Liver Function Tests Guidance
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This is to assist GPs in decision making and is not intended to replace clinical judgment.

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Consider doing LFTS
- if sx of liver/bile system disease e.g. abdo pain/nausea/vomiting/jaundice
- pt drinks excessively
- pt taking medication that affects the liver
- pt has diabetes
- Obesity
- GGT – useful in cholestasis or monitoring changes in alcohol consumption

GP Receives Abnormal LFTs:
History and Examination with attention to Alcohol consumption, Metabolic Syndrome, BMI, Hepatotoxic Drugs & Risk factors for Viral Hepatitis

Pattern Recognition

Jaundice (Bil>40) Abnormal LFTs → **Urgent Ultrasound and/or Urgent 2 week referral to secondary care or admission**

Isolated Raised Bilirubin with other normal LFTs → **Most commonly due to Gilbert's syndrome (unconjugated hyperbilirubinaemia - affects 5% of the population and is benign)**

Less commonly due to haemolysis (Consider Reticulocyte count, LDH, haptoglobin) → **Repeat LFTs, FBC, Retic Ct.: if Gilbert's confirmed then inform patient and provide information**

Normal Bilirubin with Hepatitic LFTs (ALT>ALP)

- **ALT<300 IU/L** → **Repeat within one month with AST ,GGT, FBC to confirm still elevated**

  Consider HCV and HBV

  If alcohol consumption >21U/Week (14U/Week proposed) ♂ , >14U/ Week ♀ then consider referral to alcohol services

  If abnormal → **Ultrasound & Liver Test Panel:**
  - Hepatitis B & C
  - Autoantibodies
  - Ferritin / Transferrin satn
  - Caeruloplasmin
  - Immunoglobulins
  - A1 antitrypsin

  Isolated raised LFTs but normal USS and Database → **Manage in Primary Care: Lifestyle advice and repeat LFTs in 1 year**

  Abnormal USS appearances and/or Abnormal Liver Test Panel → **Refer to Liver Specialist for possible:**
  - Viral Hepatitis
  - ALD with Advanced Fibrosis
  - PSC, PBC, Autoimmune Hepatitis
  - Gallstone disease
  - Hepatic Vascular Disorders
  - Hepatic Metabolic Disorders

  Fatty Liver Suggested by USS and/or Liver Test Panel Negative for Other Pathology Page 2

- **ALT>300 IU/L** → **Seek telephone advice and consider urgent tests** → **Ultrasound & Liver Test Panel**

Normal Bilirubin with Cholestatic LFTs (ALP>ALT) → **Repeat within one month to confirm still elevated; liver aetiology suggested by elevated GGT (otherwise consider bone aetiology and check Vitamin D)** → **Ultrasound**

- USS normal → **Ultrasound & Liver Test Panel**
- USS abnormal → **Refer to Liver Specialist for possible:**
  - Viral Hepatitis
  - ALD with Advanced Fibrosis
  - PSC, PBC, Autoimmune Hepatitis
  - Gallstone disease
  - Hepatic Vascular Disorders
  - Hepatic Metabolic Disorders

NB urgent referral if Hepatic or Biliary Malignancy suspected

Consider urgent referral pathway for suspected Hepatic and Biliary Malignancy

Clinical contact for this pathway: Prof William Rosenberg william.rosenberg@nhs.uk

Pathway Created Dec 2013 Reviewed Feb 2016
Next Review Due Feb 2019

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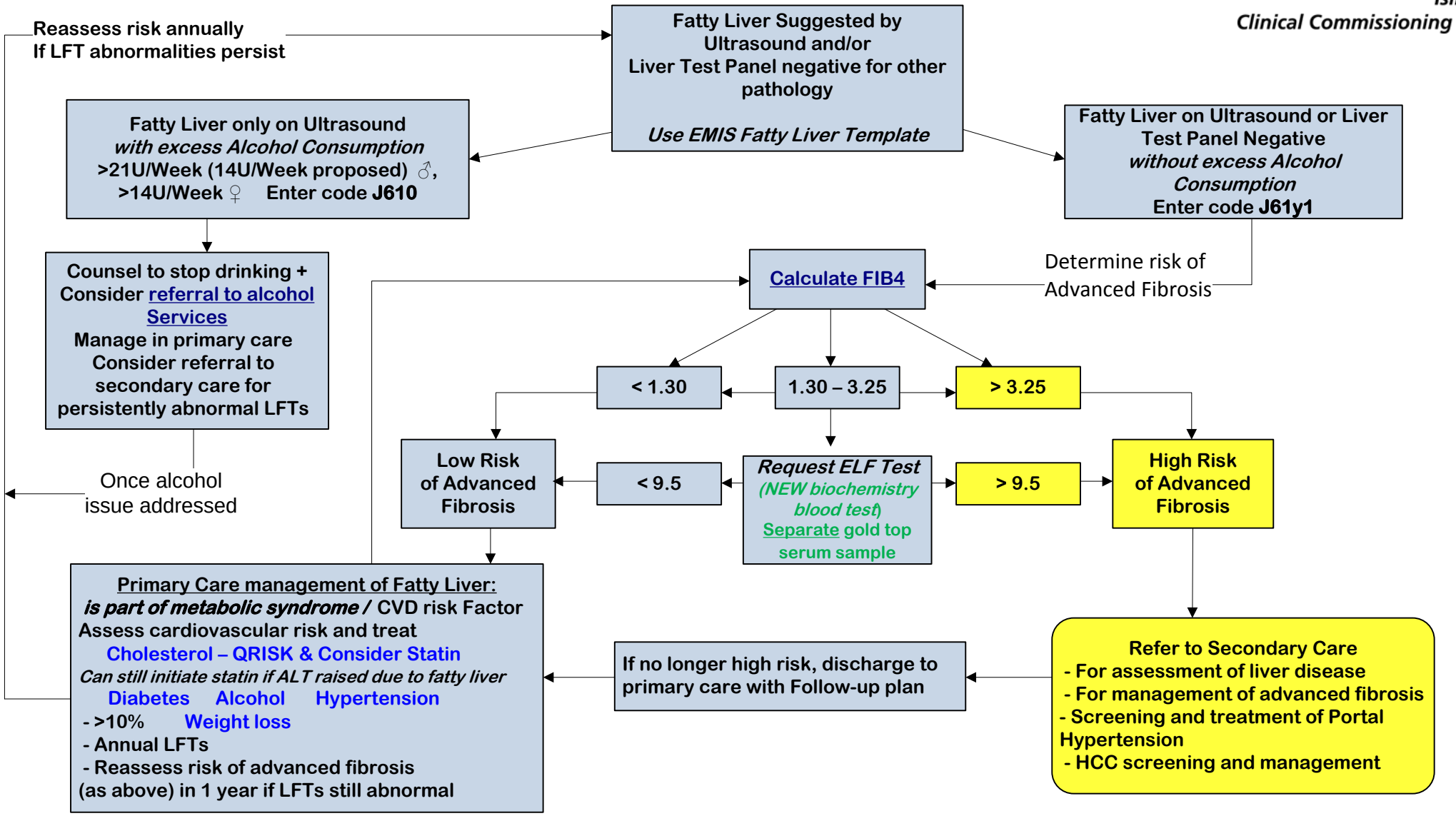

References
Fibrosis stratification in NAFLD based upon:
FIB-4 *McPherson S et al. Gut. 2010 Sep;59(9):1265-9.* FIB4: (age [yr]x AST [U/L]) / ((PLT [10^9/L]) x (√ALT [U/L]))
ELF: Enhanced Liver Fibrosis Test *Rosenberg et al. Gastroenterology. 2004 Dec;127(6):1704-13.*
Combining ELF and FIB4 in NAFLD *Tanwar et al. HEPATOLOGY. 2012, 56*, 264A.

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Primary Care Management of raised serum Ferritin.

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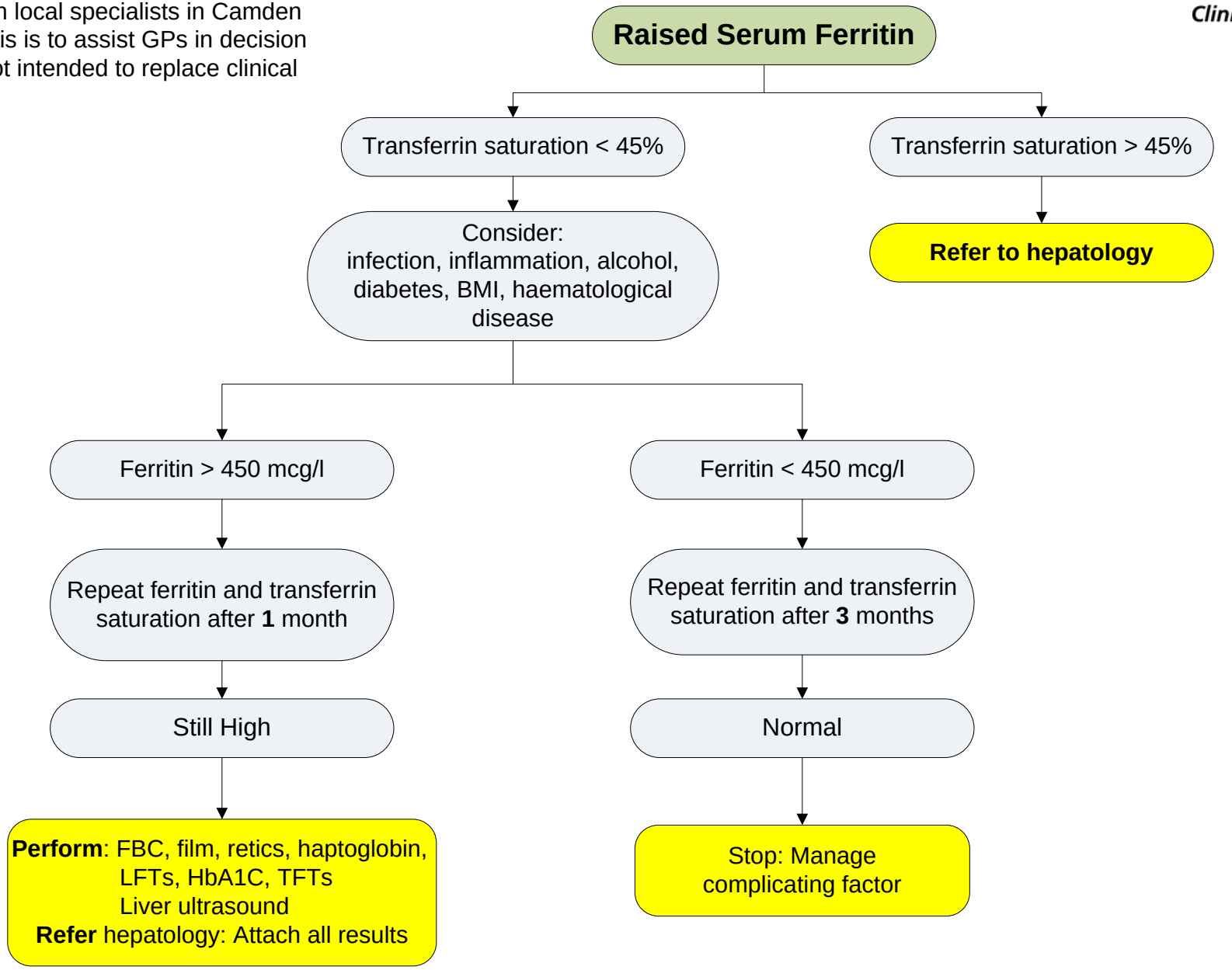

Pathway Created Dec 2013 Reviewed Feb 2016
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Primary Care Management of elevated serum GGT. This guidance has been developed in collaboration with local specialists in Camden and Islington. This is to assist GPs in decision making and is not intended to replace clinical judgment.

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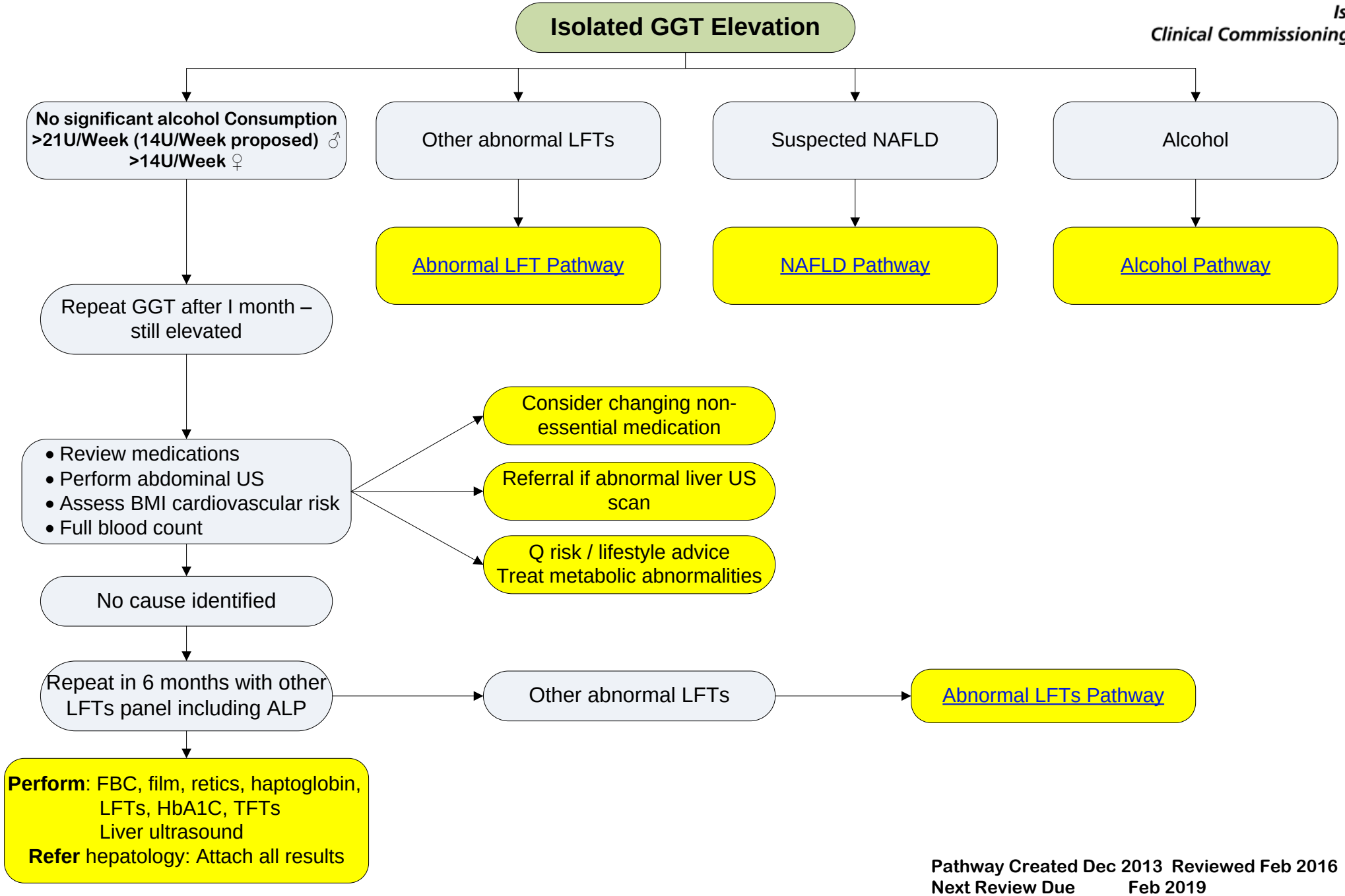

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